EV Market Insights

Project Report · Moksh Kothari · Medicaps University · Oct 29, 2026

1. Introduction

1.1 Project Overview

EV Market Insights turns 200,000 electric-vehicle records into an interactive dashboard and a guided market story that open in any web browser. It rebuilds two Tableau workbooks, the EV Executive Dashboard and the Market Story, as a Flask web app and as a static website.

The dataset covers 200 manufacturers, 17,980 models, 15 vehicle categories, 7 market regions, 10 manufacturing countries and model years 2018 to 2026. Together the records add up to 1.55 billion annual sales units. The full dataset has 36 attributes per vehicle, including battery capacity, range, charging speed, price, safety rating and customer rating. The web app uses the 11 that matter most for market analysis.

The GitHub repository holds the two Tableau workbooks, the compressed datasets, the Flask dashboard app and the static website.

1.2 Purpose

The project exists to let a reader answer EV market questions in seconds instead of reading raw tables. Its goals are:

- Show market size, average rating, average range and average battery size as six headline figures.
- Let users filter every chart at once by year, region, vehicle category, battery type, manufacturing country and manufacturer.
- Tell the main findings as a short, data-driven story whose numbers are computed from the data.
- Make the analysis shareable as a link, with no Tableau installation needed to view it.

The source table appears to be generated rather than real registration data, because totals are almost identical across regions, years and countries. The findings therefore demonstrate the analysis method and are not market facts.

2. Ideation Phase

2.1 Problem Statement

A 200,000-row EV dataset with 36 columns is too large to read by hand, and a Tableau workbook can only be opened by people who have Tableau. Analysts, students and planners need a fast way to see which regions, vehicle categories, battery types and manufacturers drive the EV market.

How can a large EV dataset be turned into a filterable, shareable view that answers market questions in seconds?

2.2 Empathy Map Canvas

The canvas below describes the main user, an analyst or student who explores the EV market to prepare a report or a decision.

Quadrant	What the user shows
Says	"Which region sells the most EVs?" "Does battery type change range or rating?"
Thinks	The raw data is too big to scan. I need totals and comparisons, not 200,000 rows.
Does	Opens the dashboard, clicks a bar or map bubble, compares vehicle categories and manufacturers, copies the link to share a view.
Feels	Frustrated by static tables and by tools that need installation. Confident when every chart updates together.

2.3 Brainstorming

Four ways to present the data were weighed, and the three that worked best were combined.

Idea	Decision
Keep the analysis only in Tableau workbooks	Kept as the starting point, but viewers need Tableau
Cross-filtering dashboard with KPI tiles, a map and seven charts	Built

Idea	Decision
Guided market story with numbers computed from the data	Built
Run as a Flask web app, or as a static site hosted for free	Both delivered

Features chosen during brainstorming: click any bar or map bubble to filter all charts, click again to clear, dropdown filters, a battery gauge showing how much of the market the current filters cover, and filters stored in the page link so any view can be shared.

3. Requirement Analysis

3.1 Customer Journey Map

A user moves from opening the dashboard to sharing a filtered view in five steps.

Stage	User action	Touchpoint	Outcome
Open	Visits the dashboard page	Six KPI tiles and battery gauge	Sees market size, rating, range and battery size at once
Explore	Reads the map and seven charts	Country map, sales, battery, manufacturer and category charts	Spots the biggest regions, categories and battery types
Filter	Clicks a bar or bubble, or uses a dropdown	Cross-filtering across all charts	Narrows the market to one slice, such as Europe and LFP
Understand	Opens the market story page	Guided narrative with computed numbers	Gets the main findings in plain language
Share	Copies the page link	Filters stored in the URL	Colleagues open the same filtered view

3.2 Solution Requirements

The solution must meet the functional and non-functional requirements below.

Functional

- Show six KPIs: manufacturers, EV models, units sold, average rating, average range, average battery size.

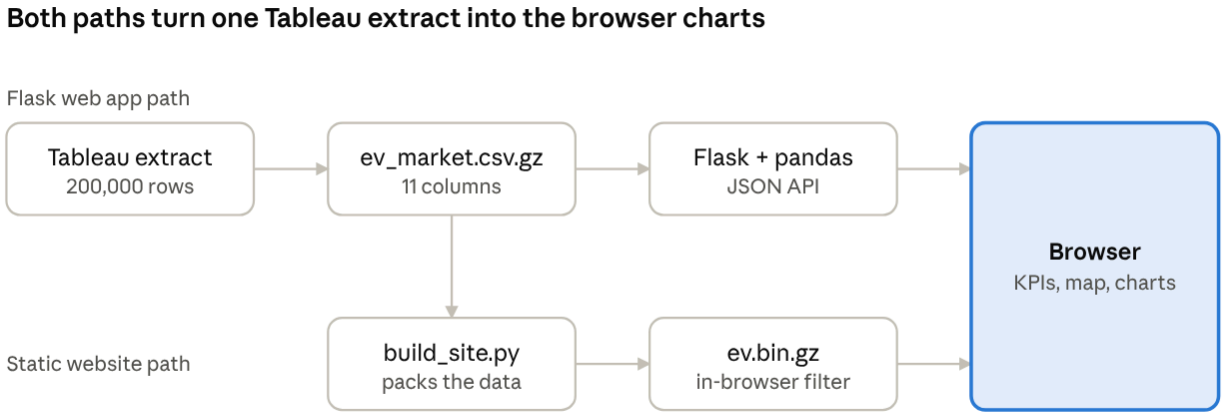
- Show a manufacturing-country map and charts for sales by region, battery type, manufacturer and category, rating by manufacturer, and range by category.
- Filter every chart together by year, region, category, battery type, country and manufacturer, with multi-select.
- Show a battery gauge for the share of the whole market covered by the current filters.
- Provide a market story page whose figures are computed from the data when the page loads.
- Keep filters in the URL so a view can be shared.
- Expose the data through a JSON API with a health check.

Non-functional

- Respond to each filter click within a few milliseconds.
- Run offline from bundled libraries, fonts and map data, with no CDN.
- Run on Python 3.10 or newer, or as a plain static site with no server.
- Deploy on free hosting.

3.3 Data Flow Diagram

Data moves from the Tableau extract, through a cleaned 11-column file, to the charts in the browser.



data flow · 2 paths, 1 shared source

The Flask path filters the data on the server; the static path packs it once and filters it inside the browser.

3.4 Technology Stack

Layer	Technology	Role
Analysis	Tableau (two .twbx workbooks)	Original dashboard and market story
Data	CSV compressed with gzip; 200,000 rows and 11 columns in the app	Source records
Backend	Python, Flask 3, pandas, NumPy	Loads data, filters, aggregates, serves the JSON API
Server	gunicorn with a Procfile	Runs the app on Render, Railway or Fly.io
Frontend	HTML, CSS, JavaScript, Chart.js, D3, topojson-client	Charts, map and cross-filtering
Static build	JavaScript query engine, packed dataset	Filters all records in the browser without a server
Design	Bricolage Grotesque and IBM Plex Sans fonts	Page typography
Testing	pytest	Checks pages, API and KPI values
Version control	Git and GitHub	Stores the project

4. Project Design

4.1 Problem Solution Fit

Each problem identified in the ideation phase has a matching feature in the solution.

Problem	Solution feature
200,000 rows are too large to read	Six KPIs and seven summary charts
Hard to compare slices of the market	Click-to-filter across every chart, plus dropdowns
Unclear how much of the market a filter covers	Battery gauge showing the share of the whole market
Findings need a written explanation	Market story page with computed numbers

Problem	Solution feature
Views cannot be passed to others	Filters stored in the page URL
Tableau is needed to open the analysis	Web app and static site that open in any browser

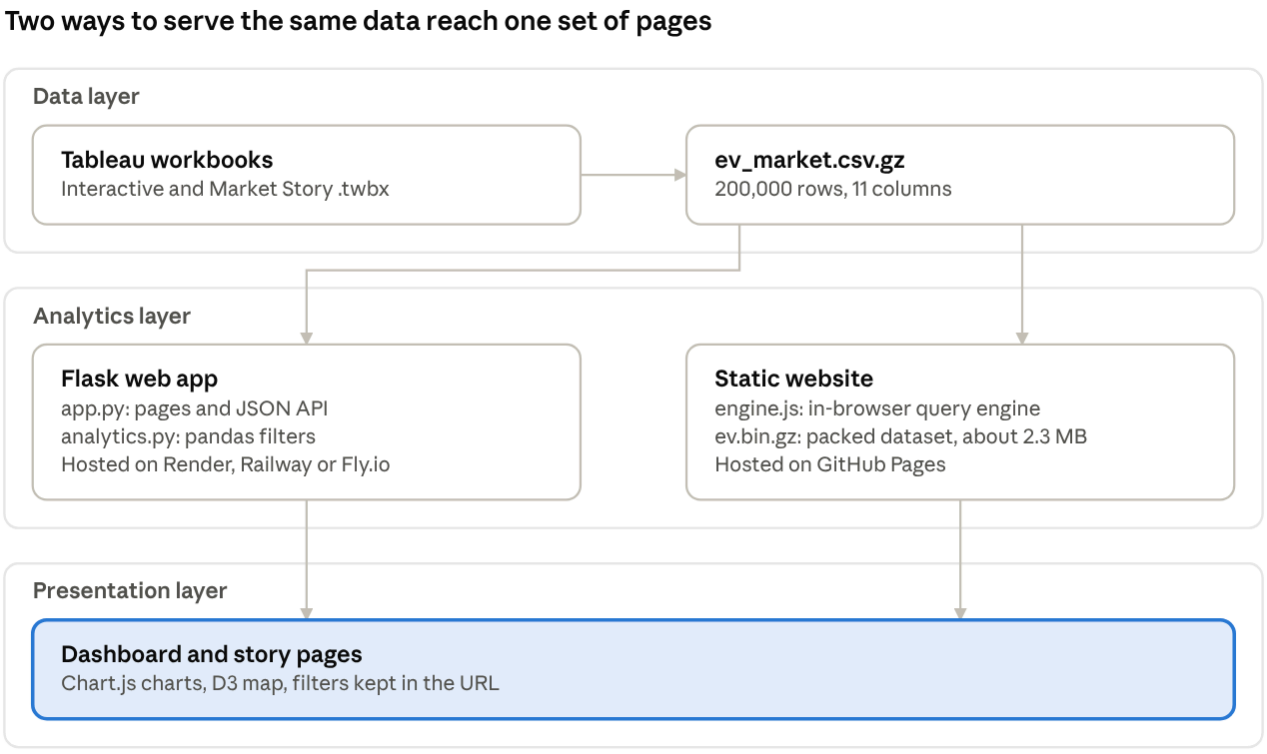
4.2 Proposed Solution

The proposed solution is a two-page web product built from the same dataset and delivered in two forms.

- **Dashboard page.** Rebuilds the Tableau EV Executive Dashboard with KPIs, a battery gauge, a manufacturing-country map and seven charts that filter each other.
- **Market story page.** A guided narrative whose text and numbers are generated from the data.
- **Flask web app.** Python and pandas aggregate the data on the server and answer JSON requests, so it needs a host that runs Python.
- **Static website.** The packed dataset of 200,000 records, about 2.3 MB, is downloaded once and filtered in the browser, so it can be hosted for free on GitHub Pages.

4.3 Solution Architecture

The architecture has a data layer, an analytics layer and a presentation layer, with two ways to serve it.



architecture · 3 layers, 2 serving options

Both options read the same cleaned data and feed the same dashboard and story pages.

5. Project Planning & Scheduling

5.1 Project Planning

The project ran in six phases, from understanding the data to publishing the finished site.

Phase	Activities	Deliverable
1. Data understanding	Studied the 36-column EV dataset and chose the 11 columns the analysis needs	Compressed datasets, 100,000 and 200,000 rows
2. Analysis in Tableau	Built the Interactive dashboard and the Market Story workbook	Two .twbx workbooks
3. Data preparation	Exported the Tableau extract to a slim file with a build script	ev_market.csv.gz, 200,000 rows by 11 columns
4. Development	Wrote the pandas analytics module, Flask routes, JSON API, pages and charts	Flask dashboard app

Phase	Activities	Deliverable
5. Static version	Packed the data and wrote an in-browser query engine	Static website for GitHub Pages
6. Testing and release	Wrote pytest checks and the README with hosting steps	Tested app, documentation, GitHub repository

6. Functional and Performance Testing

All 9 automated pytest checks in the repository pass, and a dashboard query over the full 200,000 records takes about 35 to 40 milliseconds on the server.

6.1 Performance Testing

Functional tests. The pytest suite checks the app through Flask's test client.

Test	What it checks	Result
Pages render	The dashboard and story pages return HTTP 200	Pass
Health check	/healthz returns status ok	Pass
KPIs match Tableau	200 manufacturers, 17,980 models, 1.55B units, rating 4.09, range 324.5 km, battery 78.2 kWh	Pass
Filters narrow results	Region plus battery filter gives fewer sales than region alone, which gives fewer than all	Pass
Multi-select adds up	Sales for 2020 and 2021 together equal the two years separately	Pass
Unknown filters ignored	An invalid region or parameter leaves all 200,000 records	Pass
Chart ignores its own filter	The region chart still shows all 7 regions with Europe marked as selected	Pass
Selected manufacturer shown	A chosen manufacturer always appears in the manufacturer chart	Pass
Empty selection	A filter combination with no records returns 0 records and no rating	Pass

Performance measurements. The figures below were measured on one run in the analysis environment, so absolute times will differ by machine.

Operation	Time
Load the 200,000-row dataset at startup	0.41 s
Dashboard aggregation, no filter	37.5 ms
Dashboard aggregation, region and battery filters	35.4 ms
Market story aggregation	59.0 ms

The static website needs no server. It downloads the packed dataset once, about 2.3 MB, and filters it in the browser with a single pass over the 200,000 rows.

7. Results

The analysis shows a market dominated by small vehicles and two battery chemistries, with almost no difference between regions, battery types or years. The figures below are computed from the 200,000-record dataset.

- **Market size:** 1.55 billion annual sales units across 200 manufacturers and 17,980 models, with an average customer rating of 4.09 out of 5.
- **Vehicle categories:** Two-wheeler scooters lead with 174.6 million units (11.3% of the total). Electric trucks, pickup trucks and buses each sell under 13 million.
- **Battery types:** NMC (524.1 million) and LFP (496.1 million) together hold about 66% of units. Sodium-ion has 153.3 million and solid-state 110.0 million.
- **Regions:** Totals sit between 217.5 million (Europe) and 224.0 million (South Asia), a gap of about 3%.
- **Range:** Luxury sedans average 550.6 km and luxury SUVs 524.8 km, while scooters average 105.0 km. Range and rating barely change by battery type: 324.1 to 325.9 km and 4.08 to 4.10.
- **Manufacturers:** The largest, AmpereWorks, sells 9.5 million units, about 0.6% of the market, so no manufacturer dominates.

7.1 Output Screenshots

Figure 1. Dashboard with no filters. The whole market of 1.55 billion units, six KPIs, the manufacturing-country map and seven charts.

EV Market Performance

Dashboard | Market story

EV market insights

Sales, mix and global reach across 200,000 vehicle records. Click any bar or map bubble to filter every chart. Click it again to clear.

Model year: All | Market region: All | Category: All | Battery type: All | Manufacturing country: All | Reset filters

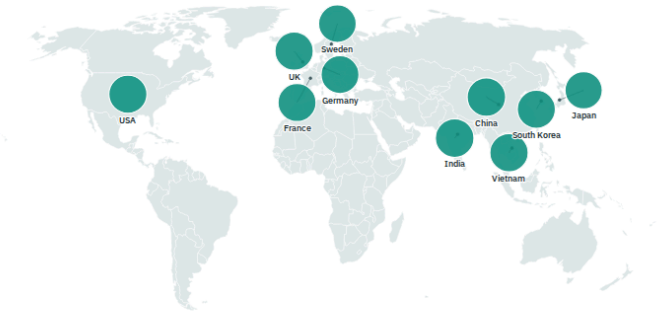
No filters applied. Showing the whole market.

100% Whole market in view: 1.55B units

200 Manufacturers	**17,980** EV models	**1.55B** Units sold
4.09 / 5 Average customer rating	**324.5** km Average range	**78.2** kWh Average battery size

Where vehicles are made

Units sold by manufacturing country. Bubble area shows sales.



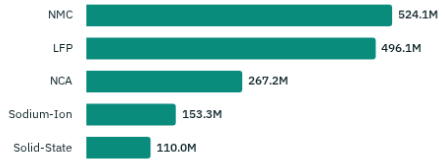
Sales by market region

Units sold in each region.



Sales by battery type

Units sold by battery chemistry.



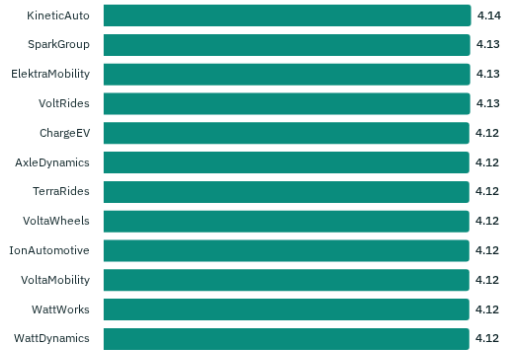
Top manufacturers by sales

Units sold, 12 largest manufacturers.



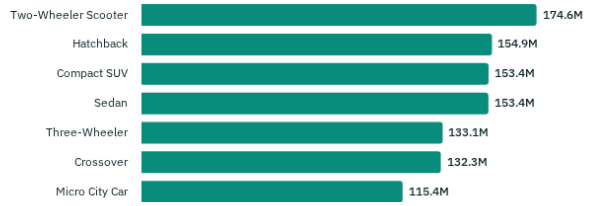
Customer rating by manufacturer

Average rating out of 5, 12 highest-rated manufacturers.



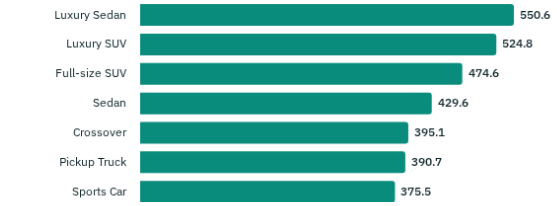
Sales by vehicle category

Units sold in each of the 15 categories.



Average range by category

Kilometres on a full charge.



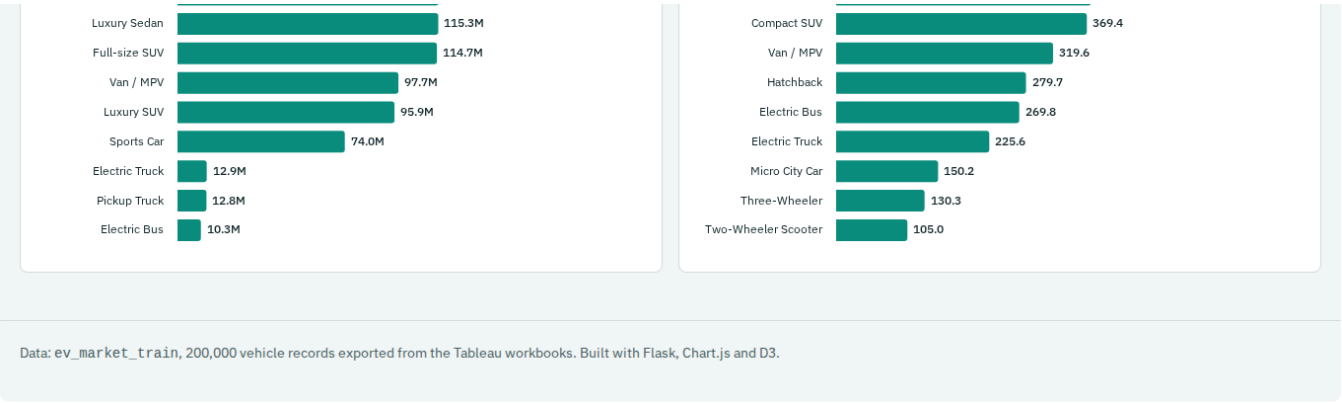
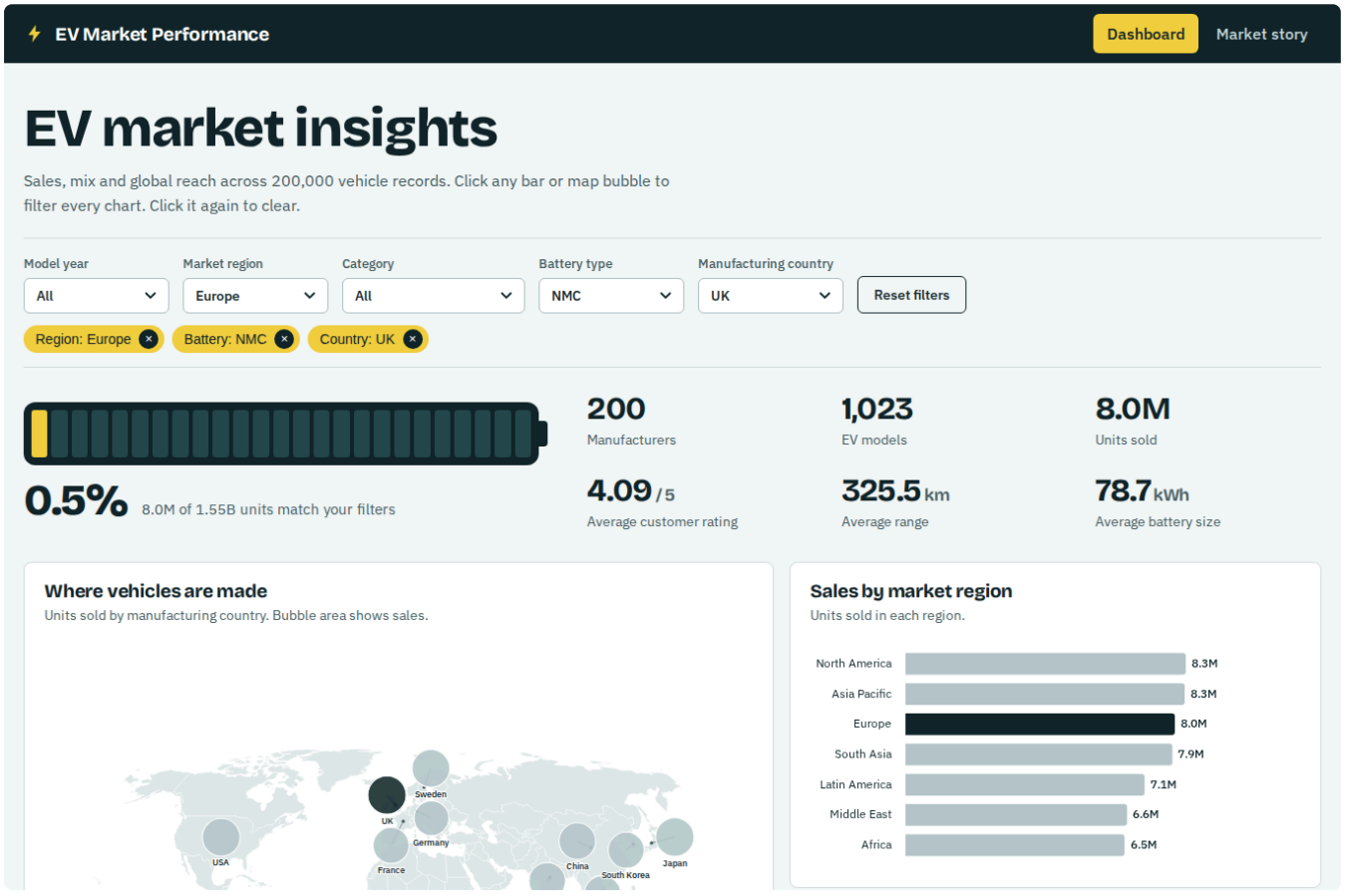


Figure 2. Dashboard filtered to Europe, NMC batteries and UK manufacturing. The battery gauge falls to 0.5% of the market, 8.0 million units, and every chart updates together.



8. Advantages & Disadvantages

The project makes a large dataset easy to explore, but its conclusions are limited by the data it is built on.

Advantages	Disadvantages
Every chart updates together from one click	The data looks generated, so findings are not real market facts

Advantages	Disadvantages
Filters live in the URL, so any view is shareable	The Flask version needs a host that runs Python; GitHub Pages cannot run it
Opens in any browser, with no Tableau needed	The web app uses 11 of the 36 available attributes
Static version works with no server and costs nothing to host	The static version needs a modern browser with gzip support
Libraries, fonts and map are bundled, so it works without a CDN	The dashboard shows past data only, with no forecasting
Automated tests confirm the numbers match the Tableau dashboard	The Tableau workbooks are large, about 10.9 MB each

9. Conclusion

EV Market Insights turns 200,000 EV records into a shareable dashboard and market story that answer common market questions in seconds. The analysis shows a market led by two-wheeler scooters and by NMC and LFP batteries, with similar totals across regions and no dominant manufacturer. Automated tests confirm that the web app reproduces the Tableau figures, and the project runs both as a Flask app and as a free static site. Because the dataset appears to be generated, the project is best read as a working demonstration of an end-to-end analytics workflow, from Tableau analysis to a deployed web product.

10. Future Scope

The project can grow in four directions, chiefly by replacing the generated data with real registration data.

- **Real data.** Load actual EV registrations and sales from official sources, so the findings describe the real market.
- **More attributes.** Use the unused columns, such as price, charging speed, safety rating, resale value and CO2 saved, to compare cost, charging and sustainability.
- **Forecasting.** Add trend lines and models that project sales by region and vehicle category.
- **Wider reach.** Add data export, user accounts for saved views, and a mobile-friendly layout, then host the Flask app on a cloud platform.

11. Appendix

Source Code

The repository contains the code in two zip files and the workbooks and data beside them.

Item	Contents
ev-market-dashboard (1).zip	Flask app: app.py, analytics.py, templates, static files, scripts/build_data.py, tests/test_app.py, Procfile
ev-market-website.zip	Static site: index.html, story.html, assets, tools/build_site.py, tools/analytics.py
EV_Market_Performance_Interactive_Refreshed[1].twbx	Tableau interactive dashboard workbook
EV_Market_Performance_Market_Story_Refreshed[1].twbx	Tableau market story workbook

Dataset Link

The datasets are stored in the GitHub repository: compressed_data.csv.gz with 100,000 rows and compressed_data-2.csv.gz with 200,000 rows, each with 36 columns. The source table in Tableau is named ev_market_train.

GitHub & Project Demo Link

- GitHub: github.com/Moksh1328/EV-market-insights
- Project demo: add the hosted link here once the site is published.